05 Linux File System and Permissions

1. What is Linux?

Linux is an open-source operating system that manages computer hardware, software, files, memory and processes.

It allows users to interact with a computer using commands or a graphical user interface (GUI).

Main features of Linux

- **Open-source:** Its source code is publicly available.
- **Multiuser:** Multiple users can have accounts on the same system.
- **Multitasking:** It can run multiple programs at the same time.
- **Secure:** It provides users, groups and file permissions.
- **Portable:** It runs on computers, servers, virtual machines and embedded devices.
- **Command-line support:** Users can control the system through a terminal.

Examples of Linux distributions

- Ubuntu
- Kali Linux
- Debian
- Fedora
- Linux Mint

Uses of Linux in cybersecurity

- Kali Linux is commonly used for penetration testing.
- Ubuntu is useful for learning Linux administration and networking.
- Linux servers are used to host websites and applications.
- Security professionals use Linux tools for log analysis, network monitoring and incident investigation.

Remember: Linux is an operating system widely used in servers, cloud computing and cybersecurity.

2. Linux File System Hierarchy

The Linux file system organizes files and directories in a tree-like structure.

Unlike Windows, which commonly uses drive letters such as C: and D:, Linux uses a single root directory represented by /.

Important Linux directories

Directory	Purpose	Example
/	Root directory; the starting point of the entire file system	/home/user
/etc	Stores system-wide configuration files	/etc/hosts
/home	Contains regular users' personal directories	/home/student
/root	Home directory of the root administrator	/root
/var	Stores frequently changing data	/var/log
/var/log	Contains system and application logs	/var/log/syslog
/tmp	Stores temporary files	/tmp/test.txt
/bin	Contains essential user commands on many Linux systems	/bin/ls
/usr	Contains many applications, libraries and shared resources	/usr/bin
/dev	Contains device files	/dev/null
/proc	Provides information about running processes and the kernel	/proc/cpuinfo
/mnt	Commonly used as a temporary mount point	/mnt/backup
/opt	Often used for additional software packages	/opt/application

Important points

- / is the root of the file system.
- /root is the root user's home directory.
- /home contains personal directories for regular users.
- /etc contains configuration files.
- /var/log is especially important in digital forensics and incident response.
- /tmp contains temporary files and may be cleaned automatically.

Example commands

ls /

ls /etc

ls /home

ls /var/log

These commands list the contents of the specified directories.

3. Linux Users and Groups

Linux supports multiple users. Each user has an identity and specific permissions.

A **user** is an account that can log in and access resources. A **group** is a collection of users used to manage permissions more easily.

Types of users

User type	Description
Root user	Administrator with extensive privileges
Regular user	Used for normal daily activities
System user	Usually created to run system services or applications

Why are groups important?

Suppose three students need access to a shared project folder. Instead of granting permissions separately to each person, an administrator can create one group and give that group access.

Commands to manage users and groups

Create a user:

sudo adduser student1

Creates a new user named student1 on Ubuntu.

Create a group:

sudo groupadd cybersecurity

Creates a group named cybersecurity.

Add a user to a group:

sudo usermod -aG cybersecurity student1

Adds student1 to the cybersecurity supplementary group.

Display the current user:

whoami

Display user and group IDs:

id student1

Display groups for the current user:

groups

List local user accounts:

cat /etc/passwd

Note: /etc/passwd contains account information, not users' actual passwords. Password hashes are normally stored separately in /etc/shadow, which is restricted.

4. Linux File Permissions

Linux permissions control who can read, modify or execute a file.

There are three basic permission types.

Permission	Symbol	Numeric value	Meaning
Read	r	4	View file contents or list directory entries
Write	w	2	Modify a file or create/delete directory entries, subject to directory permissions
Execute	x	1	Run an executable file or access a directory

Three permission categories

- **User (u):** The file owner.
- **Group (g):** Users belonging to the file's group.
- **Others (o):** Everyone else.

Example: Understanding permissions

Run:

ls -l

Example output:

-rwxr-xr-- 1 student student 120 Oct 20 10:00 script.sh

The permission string is:

- rwx r-x r--

| | |

| | └── Others: read

| └──────── Group: read and execute

└──────────── Owner: read, write and execute

In this example, the group has r-x, not r--.

Numeric permission calculation

The numeric values are:

- Read = 4
- Write = 2
- Execute = 1

Therefore:

Permission	Calculation	Numeric value
rwx	4 + 2 + 1	7
rw-	4 + 2	6
r-x	4 + 1	5
r--	4	4
---	0	0

For example, 755 means:

- Owner: 7 = rwx
- Group: 5 = r-x
- Others: 5 = r-x

5. The chmod Command

chmod means **change mode**. It changes file or directory permissions.

Example 1: Give the owner full permissions

chmod 700 secret.txt

Result:

- Owner: read, write and execute
- Group: no permissions
- Others: no permissions

For a regular text file, execute permission is usually unnecessary. To give the owner read and write access only, use chmod 600 secret.txt.

Example 2: Set common script permissions

chmod 755 script.sh

Result:

- Owner: read, write and execute
- Group: read and execute
- Others: read and execute

Example 3: Remove write permission from others

chmod o-w report.txt

Removes write permission from others.

Example 4: Give the owner read and write access

chmod u+rw report.txt

Adds read and write permissions for the owner.

Example 5: Change directory permissions

chmod 750 project

The owner gets full access, the group gets read and execute access, and others get no access.

Remember: chmod changes permissions; it does not change the file owner.

6. The chown Command

chown means **change owner**. It changes the ownership of a file or directory.

Example 1: Change the file owner

sudo chown student report.txt

Makes student the owner of report.txt.

Example 2: Change owner and group

sudo chown student:cybersecurity report.txt

Sets the owner to student and the group to cybersecurity.

Example 3: Change ownership of a directory and its contents

sudo chown -R student:cybersecurity project/

Changes ownership recursively for the project directory and its contents.

Use -R carefully because it affects every file and subdirectory beneath the specified directory.

chmod vs chown

Feature	**chmod**	**chown**
Full meaning	Change mode	Change owner
Main purpose	Changes permissions	Changes owner or group
Example	chmod 640 file.txt	chown student file.txt

Easy trick: chmod = permissions; chown = ownership.

7. What is SUID?

SUID stands for Set User ID. It is a special permission used mainly with executable files.

When an executable has SUID enabled, it normally runs with the effective user ID of the file owner rather than the ID of the user who started it.

Example: The passwd command

On many Linux systems, the passwd executable has SUID enabled because changing a user's password requires access to protected password-related files.

Check its permissions:

ls -l /usr/bin/passwd

On systems where SUID is configured, the output may resemble:

-rwsr-xr-x 1 root root ... /usr/bin/passwd

The s in the owner's execute position indicates SUID when the owner's execute bit is also set.

How SUID works

1. A regular user runs an SUID executable.
2. The program runs with the effective user ID of the file owner.
3. The program performs its intended privileged operation.
4. The program should restrict what actions the user can perform.

Security risks of SUID

- A vulnerable SUID program may allow privilege escalation.
- A poorly written program may perform unauthorized actions with elevated privileges.
- Unnecessary SUID permissions increase the attack surface.

Find SUID files

find / -type f -perm -4000 2>/dev/null

This searches for files with the SUID bit set. Permission errors are suppressed by 2>/dev/null.

Important: Finding an SUID file does not automatically mean it is malicious. Investigate whether the file is expected and properly configured.

8. Basic Linux Commands

The following commands are useful for daily Linux administration and cybersecurity lab work.

Command	Purpose	Example
pwd	Displays the current directory	pwd
ls	Lists files and directories	ls -la
cd	Changes directory	cd /home
mkdir	Creates a directory	mkdir project
touch	Creates an empty file if it does not exist	touch notes.txt
cat	Displays file contents	cat notes.txt
less	Views a file page by page	less /var/log/syslog
cp	Copies files	cp a.txt b.txt
mv	Moves or renames files	mv old.txt new.txt
rm	Removes files	rm test.txt
rmdir	Removes an empty directory	rmdir emptydir
find	Searches for files and directories	find /home -name "*.txt"
grep	Searches text for a pattern	grep "error" app.log
whoami	Displays the current username	whoami
id	Displays user and group IDs	id
groups	Displays group memberships	groups
chmod	Changes permissions	chmod 640 file.txt
chown	Changes ownership	sudo chown student file.txt
ps	Displays process information	ps aux
top	Monitors running processes	top
kill	Sends a signal to a process	kill 1234
df	Shows file system disk usage	df -h
du	Shows file or directory space usage	du -sh project/
free	Displays memory usage	free -h

ip a	Displays network interfaces and IP addresses	ip a
ping	Tests network reachability	ping -c 4 127.0.0.1
ss	Displays socket and network connection information	ss -tulpn
sudo	Runs an authorized command with elevated privileges	sudo apt update
man	Opens a command's manual	man chmod
history	Shows shell command history	history

Important command explanations

1. find

find /home -name "notes.txt"

Searches the /home directory and its subdirectories for a file named notes.txt.

2. grep

grep -i "failed" auth.log

Searches for the word failed, ignoring letter case.

3. ps

ps aux

Displays information about running processes.

4. ss

ss -tulpn

Shows TCP and UDP sockets, listening ports and associated process information when permissions allow.

5. ip a

ip a

Displays network interfaces and their assigned IP addresses.

9. Quick Revision: Important Differences

Term	Simple meaning
Linux	An open-source operating system kernel; Linux distributions combine it with other software to form a complete OS
Root directory /	The starting point of the Linux file system
Root user	The administrator account with extensive privileges
/etc	System configuration files
/var/log	System and application logs
/home	Regular users' home directories
chmod	Changes file permissions
chown	Changes file owner or group
SUID	Allows an executable to run with the effective user ID of its owner
rwx	Read, write and execute
find	Searches for files and directories
grep	Searches text for matching patterns
ps	Displays process information
ss	Displays socket and network connection information

10. Conclusion

Linux is an important foundation for cybersecurity because it helps professionals manage systems, investigate logs, monitor processes and understand network activity.

Understanding the Linux file system, users, groups, permissions, chmod, chown and SUID helps administrators protect systems against unauthorized access and privilege escalation.

Regular practice in a safe Ubuntu virtual machine will improve command-line skills and prepare you for SOC, VAPT and Digital Forensics and Incident Response (DFIR) tasks.